

Allosteric Enzymes

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Introduction

Enzymes are proteins capable of greatly accelerating the rate of a chemical reaction. While the study of enzyme structure and function is important, the study of their regulatory mechanisms is even more crucial, as a perfect harmony exists within the organism thanks to the finely regulated action of these enzymes.

Organisms are able to modulate cellular enzymatic activity in order to adapt to metabolic changes. In this way, enzymes can coordinate metabolic pathways in response to the environment.

Some enzymes possess properties that specifically allow them to play a regulatory role in metabolism. Such highly specialized enzymes are called **regulatory enzymes**, such as **allosteric enzymes**.

I. Allosteric Regulation

I.1. Definition of an Allosteric Enzyme

Allosteric enzymes are polymeric enzymes composed of an even number of subunits, most often four (04), whose enzymatic activity varies depending on the conformation of their monomers.

- **Allo:** other
- **Stereos:** site or shape

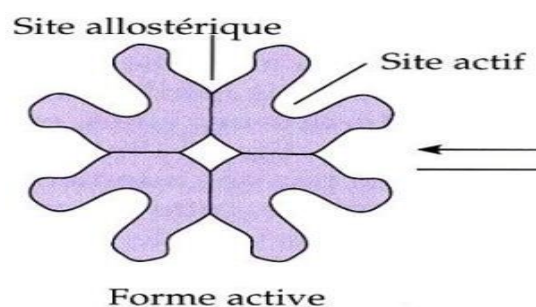


Figure 1 : Representation of an allosteric enzyme

Allostery: a property of certain active proteins that can change conformation when they bind to an allosteric effector at a site different from the active site. This binding results in a modification of their activity.

I.1. Classification

According to Monod, Wyman, and Changeux (1965), three groups of regulatory enzymes are distinguished:

1. **Homotropic enzymes:** In this group, the substrate also acts as a modulator; it accelerates enzymatic activity.
2. **Heterotropic enzymes:** These enzymes are stimulated or inhibited by regulatory substances that are unusual and different from the substrate; these are effectors.
3. **Homo-heterotropic enzymes:** In this group, the enzymes have both the substrate and other molecules as effectors.

II. Enzymes and allosteric effectors

The molecule that binds to the allosteric site is called an effector (it may also be called a modulator), and it regulates the activity of the enzyme to which it binds. The activity of the enzyme is increased when a positive allosteric effector binds to the allosteric site.

Whereas the activity of the enzyme is decreased when a negative allosteric effector binds to the allosteric site—they inhibit the enzyme.

II.1. Action of allosteric effectors

These are substances that act as activators or inhibitors by modifying the conformation of substrate-binding sites, which changes the affinity for the substrate.

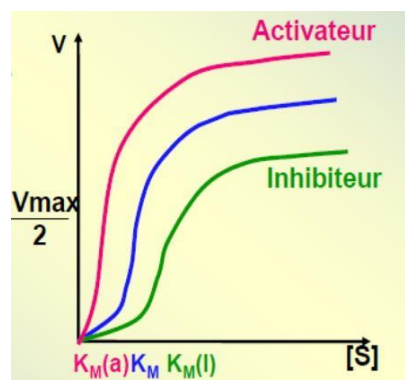


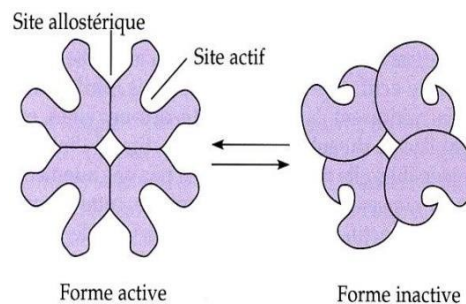
Figure 1 : Mode of action of allosteric effectors

Allosteric activators:

Increase the enzyme's affinity for the substrate, allowing it to act at low substrate concentrations.

Allosteric inhibitors:

Like competitive inhibitors, they decrease the enzyme's affinity for the substrate.



I. Regulation of enzymatic activity by the allosteric effect

The allosteric effect first arose from the following observation:

In certain enzymatic systems, the key enzyme (catalyzing the committed step in a metabolic pathway or the slowest reaction) is inhibited by the final product of this pathway = feedback regulation. Only this key enzyme = regulatory enzyme is inhibited by the final product, and it is only the final product, and it alone, that inhibits this enzyme. The product has a structure different from that of the substrate, and therefore never acts through competitive inhibition. Thus, this action of the final product is not due to an isosteric effect but rather to an allosteric effect. This final product, endowed with regulatory activity, is called an allosteric effector.

Example:

- Threonine deaminase, an enzyme located at the beginning of the isoleucine synthesis pathway, is inhibited by isoleucine.

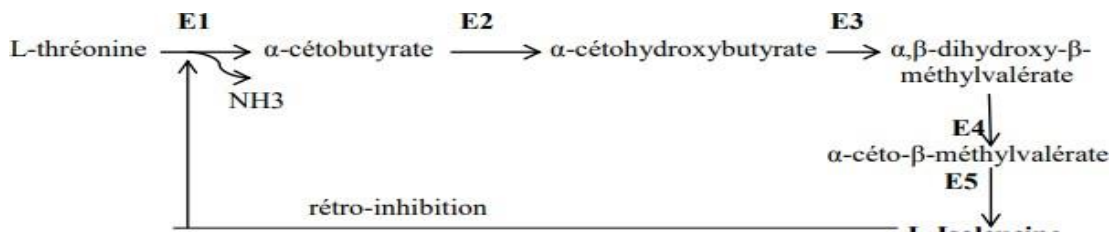


Figure2. Rétro-inhibition (feed back control) de l'enzyme thréonine-désaminase

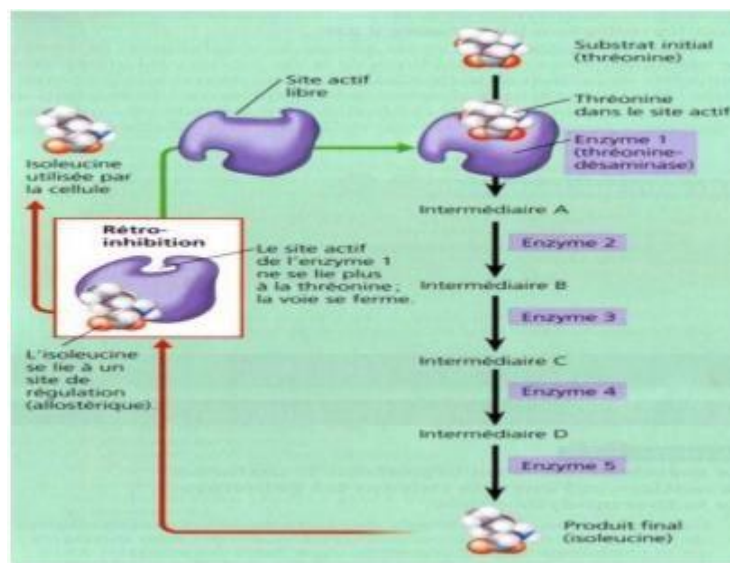


Figure 2 : Feedback regulation of allosteric enzymes

Role of allosteric controls

The role of these allosteric controls is to modulate the activity of this key enzyme according to the energy needs of the cell. Inhibitors are all indicators of a high energy charge state of the cell, whereas activators indicate a lack of energy resources.

I. Concept of cooperativity

The study of allosteric enzymes has shown that they are composed of several identical active sites and several identical allosteric sites. However, these sites do not act independently from one another; instead, they exhibit a certain cooperativity, meaning they enhance their own reaction potential through reciprocal interactions.

Cooperativity reflects the fact that the binding of a molecule (activator or inhibitor) to the enzyme influences the activity of its ligands (substrates, effectors), either by increasing or decreasing cooperativity.

I. Molecular models

Two different models have been proposed to explain the cooperative binding of ligands. They are based on the hypothesis, confirmed experimentally, that two distinct three-dimensional conformations exist for the allosteric enzyme: one active and the other inactive.

The active conformation is called relaxed (R), whereas the inactive form is called tense (T). The R form binds the substrate and the activator, while the T form binds inhibitors. Substrates and effectors are referred to as ligands of the allosteric enzyme.

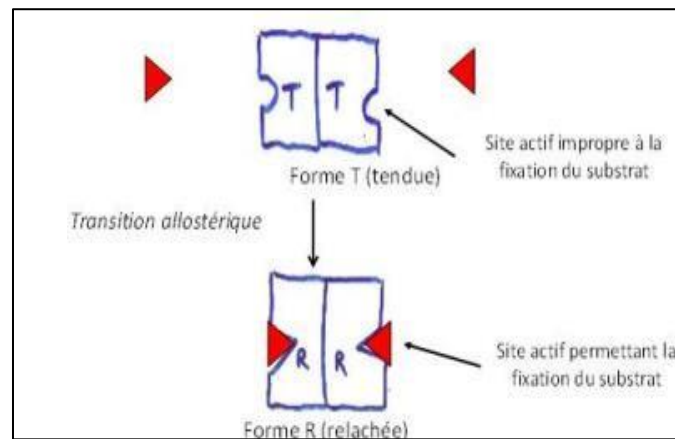


Figure 3 : Allosteric transition between the T and R states

1. Symmetric (concerted) model: Jacques Monod (1965)

An allosteric enzyme exists only in two conformations, active or inactive. All subunits are either in the active form or in the inactive form. Each substrate molecule that binds increases the probability of transition from the inactive form to the active form.

2. Sequential model: Koshland (1966)

The $T \rightarrow R$ transition does not occur all at once; the binding of a substrate to one subunit induces its transition and facilitates the transition of neighboring subunits.

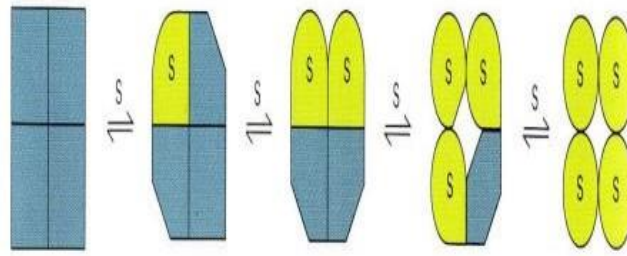


Figure 4 : Sequential binding of the ligand in the sequential model of allostery

I. Kinetic characteristics of allosteric enzymes

Allosteric enzymes do not follow classical Michaelis–Menten kinetics but instead display a sigmoidal curve, reflecting a cooperativity phenomenon. Indeed, in allosteric enzymes, the active site of one enzyme molecule can influence another active site within the same enzyme molecule.

A consequence of this interaction between subunits is that, from a kinetic point of view, cooperativity is expressed by velocity–substrate curves that are no longer hyperbolic but sigmoidal. It is as if the first substrate molecules bind with difficulty, but once bound, they facilitate the binding of subsequent ones.

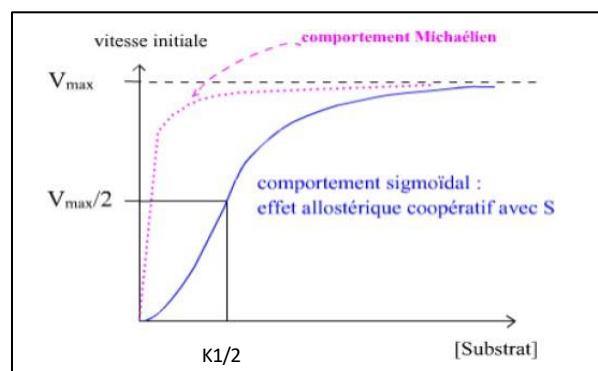


Figure 5 : Curve with Michaelian behavior for one and allosteric behavior for the other

It is no longer possible to define K_m as in Michaelian kinetics. Therefore, a purely descriptive term is used: $K_{1/2}$, which is the substrate concentration that produces half of the maximum velocity.

I. Importance of allosteric regulation

Allosteric regulation plays a very important role in cellular life. It allows the synthesis or degradation of a compound to be modulated according to its concentration, thereby adapting the cell's metabolism to its current needs. Feedback inhibition prevents the accumulation of synthesized products, while allosteric activators promote the synthesis of an important compound whose concentration becomes relatively insufficient in the cellular environment.

Conclusion

Allosteric enzymes are characterized by a sigmoidal curve (cooperativity), and their activity is modulated by effectors. All allosteric enzymes are composed of multiple subunits arranged around symmetry axes. Several lines of evidence show that the symmetric model is able to simply explain the properties of these enzymes.